QR BASED COLLEGE CANTEEN MANAGEMENT SYSTEM

Abstract

The QR Based College Canteen Web Application is an innovative platform designed to enhance College Canteen experience using modern methods like QR Code Scanning, Order ID confirmation. This Project contains Three modules 1.Admin,2.Canteen.3.Student.The Admin Module contains a unified dashboard where it displays total number of students registered and total number of canteen staff assigned and the total revenue generated through student orders in a graphical representation. The Admin can manage the canteen like add/edit/delete the canteen staff and admin can also download the reports in a Excel format, Admin can view the feedbacks that was given by Students. The Canteen Module also contains a unified dashboard where it displays the number of orders are pending and number of orders are confirmed. The Canteen can manage food items like add/delete food items and also manages inventory like updating stock of the items. Canteen can instantly retrieve an order details by entering or scanning the student's QR code or Order ID to confirmation. The Student dashboard also contains a unified dashboard that contains list of food items. Student Can add the food items into cart to place the order after payment Students get Order ID and a QR code. By providing these Order ID or QR code to the Canteen, provides the food items to Students. The Web Application is developed using HTML, CSS, JavaScript for an interactive and responsive frontend, while Django serves as Backend framework for handling authentication and data storage.

Keywords: QR Code Scanning, OTP Based Rest-Password, Food Ordering System, Canteen Inventory Management

Problem Statement

Traditional college canteen systems often rely on manual processes for food ordering, inventory tracking, and user management, leading to long queues, order mismatches, inefficient staff coordination, and lack of real-time data for decision-making. There is a need for a streamlined, digital solution that enables students to place orders efficiently, allows canteen staff to manage inventory and orders in real-time, and provides administrators with comprehensive insights through dashboards and downloadable reports. This project aims to

address these inefficiencies by developing a QR-based college canteen web application that simplifies the entire canteen management process using role-based modules for Admin, Canteen Staff, and Students.

Motivation

In many colleges, the existing canteen management system is outdated, relying heavily on manual operations such as taking orders verbally, maintaining handwritten inventory logs, and managing finances without centralized tracking. These practices often result in long wait times, order confusion, lack of transparency, and inefficient use of resources. With the growing demand for digital transformation and contactless services, there is a strong need for an automated, user-friendly system that enhances the overall canteen experience. The motivation behind this project is to leverage modern web technologies and QR code-based solutions to create a smart, efficient, and transparent canteen system that benefits students, canteen staff, and administrators alike. By introducing role-based modules and real-time data access, this system aims to bring speed, accuracy, and convenience to the daily operations of college canteens.

Objective

The objective of the QR-Based College Canteen Web Application is to develop a digital platform that simplifies and automates the canteen management process in colleges. This system aims to enhance user experience by allowing students to browse menus, place orders, and make payments online, receiving a unique QR code and Order ID for easy, contactless order pickup. It enables canteen staff to efficiently manage food items, update inventory, and process orders using QR code or Order ID confirmation. The admin module provides a centralized dashboard to monitor student registrations, staff details, revenue generation, and student feedback, with the ability to download reports in Excel format. Built using HTML, CSS, and JavaScript for the frontend and Django for backend operations, the system ensures a responsive, secure, and streamlined experience for all users involved.

Scope

The QR-Based College Canteen Web Application is designed to serve as a comprehensive digital solution for managing all aspects of a college canteen. The system covers three main user roles: Admin, Canteen Staff, and Students. It allows students to view available food items, place orders, make payments, and retrieve their orders using a generated QR code or Order ID.

Canteen staff can manage the menu, track and update inventory, and confirm or deliver orders through the scanning of QR codes. The admin has access to a centralized dashboard for managing staff, monitoring student registrations, analysing revenue reports, and reviewing student feedback. Reports can be exported in Excel format for administrative use. The application is built with modern web technologies, using HTML, CSS, and JavaScript for the frontend and Django for backend processing and data management. The system is scalable and can be implemented in any educational institution looking to modernize its canteen operations with enhanced efficiency, traceability, and user satisfaction.

Existing System

In many colleges, the existing canteen management system is largely manual and inefficient. Students typically stand in long queues to place their orders, make payments, and wait for their food, which leads to crowding, delays, and confusion. There is minimal to no automation in handling orders, tracking inventory, or maintaining sales data. Canteen staff manage inventory and food items manually, which often results in stock mismanagement and lack of real-time updates. Additionally, administrators have limited visibility into canteen operations, with no centralized platform to monitor revenue, student activity, or staff performance. There is also no standardized system for collecting or analysing student feedback. Overall, the traditional system lacks convenience, scalability, and real-time insights, making it difficult to manage high volumes efficiently or improve the canteen experience for users.

Disadvantages

1. Long waiting times due to manual order placement and payment
2. Crowding and inefficiency during peak hours
3. High risk of order mismatches or lost orders
4. Manual inventory management leading to stock inaccuracies
5. Inefficient reporting and data management

Proposed System

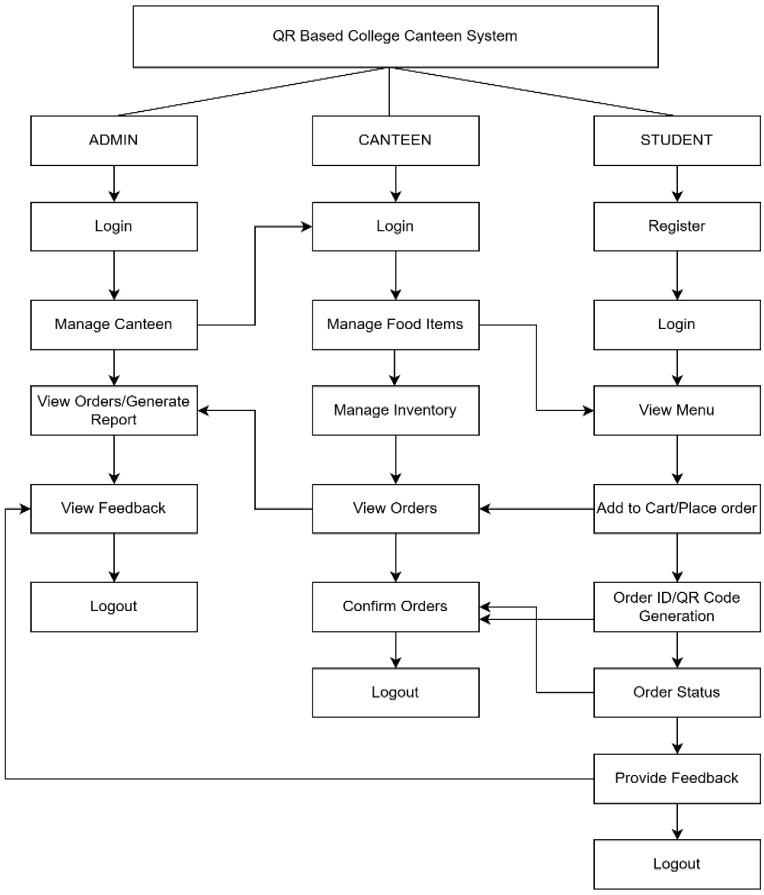
The proposed QR-Based College Canteen Web Application is a modern, digital platform designed to enhance the overall canteen experience by integrating QR code scanning and Order ID confirmation. It consists of three main modules: Admin, Canteen, and Student. The Admin

module provides a unified dashboard to monitor student registrations, canteen staff assignments, and revenue generation, along with tools for staff management, feedback review, and report downloads in Excel format. The Canteen module features a dashboard showing pending and confirmed orders, enabling staff to manage food items and inventory efficiently, while allowing quick order retrieval through QR code or Order ID scanning. The Student module offers an interactive dashboard where students can browse food items, add them to a cart, place orders, and receive an Order ID and QR code for seamless order collection. Built using HTML, CSS, and JavaScript for an interactive frontend and Django for secure backend authentication and data handling, this system streamlines order processing, improves inventory control, and enhances communication between students, canteen staff, and administrators for a smooth and efficient canteen operation.

Advantages of the Proposed System

1. Enables fast and contactless order placement using QR codes and Order IDs.
2. Reduces waiting time and crowding in the canteen.
3. Provides real-time order tracking and confirmation for canteen staff.
4. Simplifies inventory management with instant stock updates.
5. Offers a centralized admin dashboard for monitoring students, staff, orders, and revenue.

Work Flow of Proposed system



HARDWARE AND SOFTWARE REQUIREMENTS

HARDWARE REQUIREMENTS:

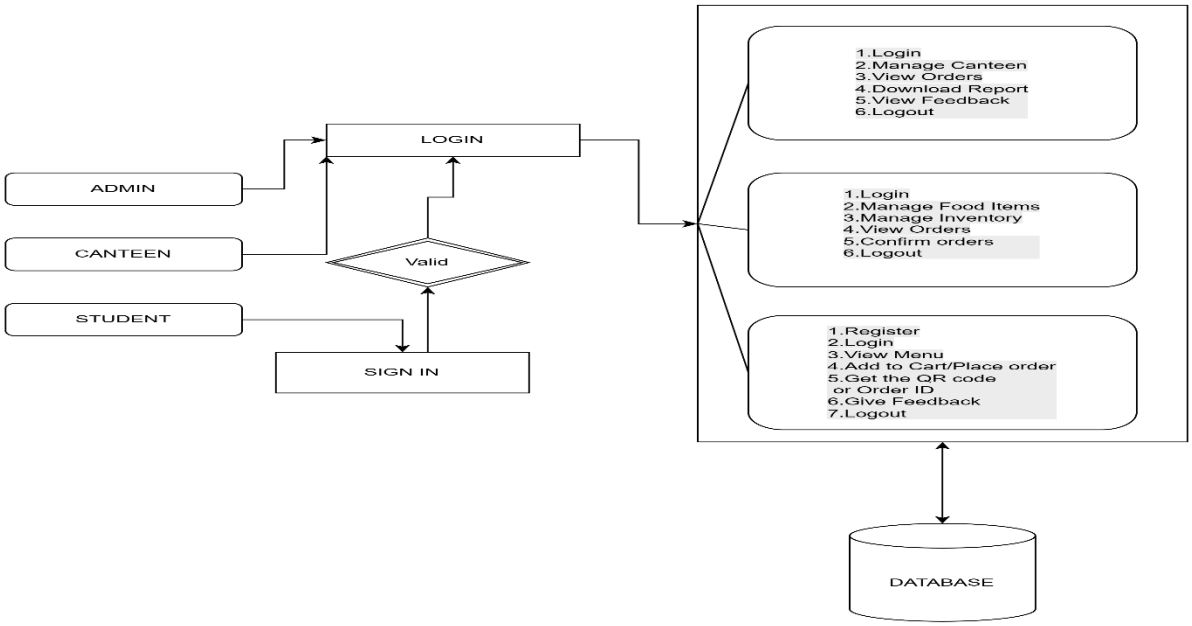
- Processor - I3/Intel Processor
- RAM - 4GB (min)
- Hard Disk - 160GB

SOFTWARE SYSTEM CONFIGURATION:

- Operating System : Windows 7/8/10
- Front-end : HTML, CSS, JavaScript
- Backend : Django
- IDE/Workbench : VS Code

- Database : SQLLITE

Architecture



Modules

1. Admin Module

- **Login:** Admin can log in securely using credentials.
- **Dashboard:** View graphical stats including:
  - Total number of registered students
  - Total canteen staff assigned
  - Total revenue generated from student orders
- **Manage Staff:**
  - Add/Edit/Delete canteen staff accounts
- **View Feedback:**
  - View feedback submitted by students about the canteen or service
- **Generate Reports:**
  - Download reports (orders, revenue, etc.) in Excel format

- **Logout:** Admin can log out after completing operations

2. Canteen Module

- **Login:** Canteen staff can log in using their credentials
- **Dashboard:**
  - View total orders
  - Track number of pending and confirmed orders
- **Manage Food Items:**
  - Add new food items to the menu
  - Delete unavailable food items
- **Inventory Management:**
  - Update stock for each food item
- **Order Confirmation:**
  - Scan student's QR code or enter Order ID to retrieve and confirm order details
- **Logout:** Canteen staff can log out after operations are complete

3. Student Module

- **Register:** Students can register by filling in their personal and college details
- **Login:** Students log in using their credentials
- **Dashboard:**
  - View the list of available food items with prices
- **Place Order:**
  - Add items to the cart
  - Make online payment
  - Receive a unique Order ID and QR Code
- **Collect Order:**
  - Present QR Code or Order ID at canteen to collect food
- **Provide Feedback:**
  - Submit feedback on canteen service or specific food items
- **Logout:** Student can log out after completing actions